Daily Algebra Drill #7 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Weighted AM–GM · Vieta jumping · Power mean inequality · Infinite products via telescoping.

Section A Rapid Recognition

1. Factorise: $x^5 - x$.

Answer: $x(x-1)(x+1)(x^2+1)$

Method: $x(x^4-1) = x(x^2-1)(x^2+1) = x(x-1)(x+1)(x^2+1)$.

Investigate further: The five roots are $0, \pm1, \pm i$. These are all fifth roots of $x^5 = x$, i.e. fixed points of $f(z) = z^5$ on $\mathbb{C}$. Separately: $x^5 - x = x(x^4-1)$. Note $x^4 - 1 = (x^2-1)(x^2+1)$. How many real roots does $x^5 - x = c$ have for various c?

2. Given $ab + bc + ca = 4$ and $a + b + c = 3$, find $(a+b+c)^2 - (a^2+b^2+c^2)$.

Answer: 8

Method: $(a+b+c)^2 - (a^2+b^2+c^2) = 2(ab+bc+ca) = 8$.

Investigate further: Instant: the identity $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$ means the answer is $2 \times 4 = 8$, regardless of the value of $a+b+c = 3$. The condition $a+b+c = 3$ is a red herring here — a classic TMUA trap.

3. Without a calculator: $7^2 - 5^2 + 3^2 - 1^2$.

Answer: 32

Method: Use $(a^2-b^2) = (a-b)(a+b)$: $(7^2-5^2) + (3^2-1^2) = (7-5)(7+5) + (3-1)(3+1) = 2 \times 12 + 2 \times 4 = 24 + 8 = 32$.

Investigate further: This telescoping of paired differences is always faster than computing individual squares.

4. Roots of $x^2 - 8x + 3 = 0$ are p, q. Find $\left(p - \frac{1}{p}\right)\left(q - \frac{1}{q}\right)$. **Method:** $\left(p - \frac{1}{p}\right)\left(q - \frac{1}{q}\right) = pq - \frac{p}{q} - \frac{q}{p} + \frac{1}{pq} = pq - \frac{p^2+q^2}{pq} + \frac{1}{pq}$. Vieta: $p + q = 8$, $pq = 3$. $p^2 + q^2 = (p+q)^2 - 2pq = 64 - 6 = 58$. Result $= 3 - \frac{58}{3} + \frac{1}{3} = 3 - \frac{57}{3} = 3 - 19 = -16$.

Answer: -16

Investigate further: Practice the general expansion: $\left(p - \frac{1}{p}\right)\left(q - \frac{1}{q}\right) = pq + \frac{1}{pq} - \frac{p^2+q^2}{pq}$. Each of pq, p^2+q^2, and $\frac{1}{pq}$ can be expressed using Vieta's formulas. The key: never try to find p and q individually.

5. Simplify: $\dfrac{(2n)!}{(2n-1)!} - \dfrac{n!}{(n-1)!}$.

Answer: n

Method: $\frac{(2n)!}{(2n-1)!} = 2n$ and $\frac{n!}{(n-1)!} = n$. Difference $= 2n - n = n$.

> **Investigate further:** Both ratios use the same identity: $\frac{k!}{(k-1)!} = k$. Memorise this as "factorial ratio = top index." Now simplify $\frac{(n+2)!}{n!} - \frac{(n+1)!}{(n-1)!}$ using the same idea.

6. Factorise: $27x^3 - 8$.

Answer: $(3x-2)(9x^2+6x+4)$

Method: Difference of cubes: $(3x)^3 - 2^3 = (3x-2)(9x^2+6x+4)$.

> **Investigate further:** The quadratic factor $9x^2+6x+4$ has discriminant $36-144 = -108 < 0$, so it is irreducible over $\mathbb{R}$. Its roots are the complex cube roots of $\frac{8}{27}$ other than $\frac{2}{3}$.

7. Given $t = x - \frac{1}{x}$, write $t^2 + 4$ in terms of x.

Answer: $\left(x + \dfrac{1}{x}\right)^2$

Method: $t^2 + 4 = \left(x - \frac{1}{x}\right)^2 + 4 = x^2 - 2 + \frac{1}{x^2} + 4 = x^2 + 2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2$.

> **Investigate further:** This identity $\left(x - \frac{1}{x}\right)^2 + 4 = \left(x + \frac{1}{x}\right)^2$ is the algebraic version of $\cosh^2 t - \sinh^2 t = 1$ (if $x = e^t$). It underpins the technique in C6 of WS#11 for moving between $x + 1/x$ and $x - 1/x$.

8. Evaluate: $\log_2 8 + \log_2 4 - \log_2 32$.

Answer: 0

Method: $3 + 2 - 5 = 0$. Convert to powers of 2: $8 = 2^3$, $4 = 2^2$, $32 = 2^5$.

> **Investigate further:** $\log_2(8\times4) - \log_2 32 = \log_2 32 - \log_2 32 = 0$. The log laws: $\log(ab) = \log a + \log b$ and $\log(a/b) = \log a - \log b$. No log rules needed: converting to powers of 2 is always faster for base-2 problems.

9. Given $f(x) = 2x+1$ and $g(x) = x^2 - 1$, find $f(g(x)) - g(f(x))$.

Answer: $-2x^2 - 4x - 1$

Method: $f(g(x)) = 2(x^2-1) + 1 = 2x^2 - 1$. $g(f(x)) = (2x+1)^2 - 1 = 4x^2 + 4x + 1 - 1 = 4x^2 + 4x$. Difference $= 2x^2 - 1 - 4x^2 - 4x = -2x^2 - 4x - 1$.

> **Investigate further:** Note $f \circ g \neq g \circ f$ in general — function composition is not commutative. Find the values of x where $f(g(x)) = g(f(x))$, i.e. where $-2x^2 - 4x - 1 = 0$. What does this mean about the relationship between the two composition paths?

10. Simplify: $\dfrac{a^2-b^2}{a-b} - \dfrac{a^2-c^2}{a-c}$. **Method:** $\frac{(a-b)(a+b)}{a-b} - \frac{(a-c)(a+c)}{a-c} = (a+b) - (a+c) = b - c$.

Answer: $b - c$

Investigate further: Both fractions simplify to sums: $(a + b)$ and $(a + c)$. The a terms cancel. This is a common TMUA trick: simplify each term individually before combining, rather than finding a common denominator.

Section B Manipulation Drills

1. Evaluate the telescoping product $\prod_{k=2}^{n}\left(1 - \frac{1}{k}\right)$ for general n and at $n = 100$.

Answer: $\dfrac{1}{n}$; at $n = 100$: $\dfrac{1}{100}$

Method: $\prod_{k=2}^{n} \frac{k-1}{k} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n-1}{n} = \frac{1}{n}$ (telescoping).

Investigate further: Compare to the sum $\sum_{k=2}^{n}\left(1 - \frac{1}{k}\right) = \sum_{k=2}^{n} \frac{k-1}{k}$, which does *not* telescope. Products telescope when successive terms are ratios; sums telescope when successive terms are differences. Know which structure you're dealing with before proceeding.

2. Simplify: $\dfrac{x^2 + 3x - 10}{x^2 - x - 2} \cdot \dfrac{x^2 + x - 2}{x^2 + 6x + 5}$. **Method:** $x^2+3x-10 = (x+5)(x-2)$, $x^2-x-2 = (x-2)(x+1)$, $x^2+x-2 = (x+2)(x-1)$, $x^2+6x+5 = (x+5)(x+1)$. So $\dfrac{(x+5)(x-2)}{(x-2)(x+1)} \cdot \dfrac{(x+2)(x-1)}{(x+5)(x+1)} = \dfrac{(x+2)(x-1)}{(x+1)^2}$.

Answer: $\dfrac{(x+2)(x-1)}{(x+1)^2}$

Investigate further: Always factor numerators and denominators completely before multiplying rational expressions — cancel at the *fraction level* before computing. This problem has 4 quadratics but only takes 4 factorisations and 4 cancellations.

3. Find $a^3 + b^3 + c^3$ given $a + b + c = 2$, $a^2 + b^2 + c^2 = 6$, $abc = -2$.

Answer: 8

Method: $ab + bc + ca = \frac{(a+b+c)^2-(a^2+b^2+c^2)}{2} = \frac{4-6}{2} = -1$. Then $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. So $a^3 + b^3 + c^3 - 3(-2) = 2(6 - (-1)) = 14$, hence $a^3 + b^3 + c^3 = 8$.

Investigate further: The two approaches: (1) Newton's identity $p_3 = e_1p_2 - e_2p_1 + 3e_3$, (2) the cubic identity $p_3 - 3e_3 = e_1(p_2 - e_2)$. Both are worth knowing. Verify with $a = 1, b = -1, c = 2$: $a + b + c = 2$ ✓, $a^2 + b^2 + c^2 = 6$ ✓, $abc = -2$ ✓, $a^3 + b^3 + c^3 = 1 - 1 + 8 = 8$ ✓.

4. Write $x^4 + 4$ as a product of two quadratics with integer coefficients.

Answer: $(x^2 + 2x + 2)(x^2 - 2x + 2)$

Method: Sophie Germain: $x^4 + 4 = (x^4 + 4x^2 + 4) - 4x^2 = (x^2 + 2)^2 - (2x)^2 = (x^2 + 2x + 2)(x^2 - 2x + 2)$.

Investigate further: Sophie Germain identity: $a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$. Here $a = x$, $b = 1$. Each quadratic factor has discriminant $4 - 8 = -4 < 0$, so neither factors further over $\mathbb{R}$. But over $\mathbb{C}$: roots of $x^2 + 2x + 2 = 0$ are $-1 \pm i$, which are primitive 4th roots of -4.

5. Find $f^{-1}(x)$ for $f(x) = \dfrac{2x+1}{x-1}$ and verify $f(f^{-1}(x)) = x$.

Answer: $f^{-1}(x) = \dfrac{x+1}{x-2}$

Method: Set $y = \frac{2x+1}{x-1}$: $y(x-1) = 2x+1 \Rightarrow xy - y = 2x+1 \Rightarrow x(y-2) = y+1 \Rightarrow x = \frac{y+1}{y-2}$. So $f^{-1}(x) = \frac{x+1}{x-2}$. Verify: $f\left(\frac{x+1}{x-2}\right) = \frac{2\cdot\frac{x+1}{x-2}+1}{\frac{x+1}{x-2}-1} = \frac{\frac{2x+2+x-2}{x-2}}{\frac{x+1-x+2}{x-2}} = \frac{3x}{3} = x$. ✓

> **Investigate further:** Note: $f^{-1}(x) = \frac{x+1}{x-2}$ has the same form as f (linear over linear). All Möbius transformations have inverses of the same type. Check: is f^{-1} also its own inverse? Compute $f^{-1}(f^{-1}(x))$. If yes, f has order 2 in the group of Möbius transformations. What is the order of f (the order of f = smallest n with $f^n(x) = x$)?

6. Prove: the product of four consecutive integers is always one less than a perfect square.

Answer: $(n^2+n-1)^2 - 1$

Method: Let the integers be $n-1, n, n+1, n+2$. Pair: $(n-1)(n+2) = n^2+n-2$ and $n(n+1) = n^2+n$. Let $m = n^2+n$: product $= (m-2)m = m^2-2m$. So product $+1 = m^2-2m+1 = (m-1)^2 = (n^2+n-1)^2$. □

> **Investigate further:** This shows $n(n-1)(n+1)(n+2) = (n^2+n-1)^2 - 1$, a perfect square minus 1. So the product is the product of (n^2+n-2) and (n^2+n) — two numbers differing by 2 whose product is always a square minus 1. Generalise: show the product of any four terms in AP is always one less than a perfect square.

7. Simplify: $\dfrac{1}{1+x} + \dfrac{1}{1-x} + \dfrac{2}{1+x^2} + \dfrac{4}{1+x^4}$.

Answer: $\dfrac{8}{1-x^8}$

Method: $\frac{1}{1+x} + \frac{1}{1-x} = \frac{2}{1-x^2}$. Then $\frac{2}{1-x^2} + \frac{2}{1+x^2} = \frac{4}{1-x^4}$. Then $\frac{4}{1-x^4} + \frac{4}{1+x^4} = \frac{8}{1-x^8}$.

> **Investigate further:** This is a *telescoping by doubling*: each step uses $\frac{1}{1-t} + \frac{1}{1+t} = \frac{2}{1-t^2}$, doubling the degree of the denominator each time. The pattern: $\sum_{k=0}^{n} \frac{2^k}{1+x^{2^k}} = \frac{2^{n+1}}{1-x^{2^{n+1}}}$ (starting from $\frac{1}{1-x}$ after the first step). This is related to the formula for the infinite product $\prod(1+x^{2^k}) = \frac{1}{1-x}$.

8. Roots α, β, γ of $x^3+px+q = 0$. Express $\alpha^3+\beta^3+\gamma^3$ in terms of p and q without Newton's identity.

Answer: $-3q$

Method: Each root satisfies the equation: $\alpha^3 = -p\alpha - q$. Similarly for β, γ. Add: $\alpha^3+\beta^3+\gamma^3 = -p(\alpha+\beta+\gamma) - 3q = -p(0) - 3q = -3q$ (since $\alpha+\beta+\gamma = 0$ for a depressed cubic).

> **Investigate further:** This elegant proof uses the polynomial equation itself to evaluate power sums. It generalises: if r satisfies $r^n + a_{n-1}r^{n-1} + \cdots + a_0 = 0$, then $r^n = -a_{n-1}r^{n-1} - \cdots - a_0$. Summing over all roots gives Newton-like identities without the recurrence formalism.

9. Prove: if $a+b+c = 0$ then $(a^2+b^2+c^2)^2 = 2(a^4+b^4+c^4)$.

Answer: Proved below.

Method: $a+b+c=0 \Rightarrow (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca) = 0 \Rightarrow ab+bc+ca = -\frac{s}{2}$ where $s = a^2+b^2+c^2$. $(ab+bc+ca)^2 = a^2b^2+b^2c^2+c^2a^2+2abc(a+b+c) = a^2b^2+b^2c^2+c^2a^2$. $(a^2+b^2+c^2)^2 = a^4+b^4+c^4+2(a^2b^2+b^2c^2+c^2a^2) = s^2$. And $(ab+bc+ca)^2 = s^2/4$, so $a^2b^2+b^2c^2+c^2a^2 = s^2/4$. Thus $a^4+b^4+c^4 = s^2 - 2(s^2/4) = s^2/2$. So $(a^2+b^2+c^2)^2 = s^2 = 2(s^2/2) = 2(a^4+b^4+c^4)$. $\square$

> **Investigate further:** This was a result from Worksheet #7 Section C. Revisiting it with the intermediate steps exposed reveals the chain of reasoning. The key chain: $a+b+c=0 \Rightarrow ab+bc+ca = -\frac{a^2+b^2+c^2}{2} \Rightarrow (ab+bc+ca)^2 = \frac{(a^2+b^2+c^2)^2}{4}$. Memorise this chain as a unit.

10. Solve: $\dfrac{x+1}{x-2}+\dfrac{x-2}{x+1} = \dfrac{10}{3}$. **Method:** Let $u = \frac{x+1}{x-2}$: $u+\frac{1}{u} = \frac{10}{3} \Rightarrow 3u^2-10u+3 = 0 \Rightarrow (3u-1)(u-3) = 0$. $u=3$: $\frac{x+1}{x-2} = 3 \Rightarrow x+1 = 3x-6 \Rightarrow x = \frac{7}{2}$. $u = \frac{1}{3}$: $\frac{x+1}{x-2} = \frac{1}{3} \Rightarrow 3x+3 = x-2 \Rightarrow x = -\frac{5}{2}$.

Check: at $x = 7/2$: $\frac{9/2}{3/2} = 3$ and $\frac{3/2}{9/2} = \frac{1}{3}$. Sum $= 3+\frac{1}{3} = \frac{10}{3}$. ✓

Answer: $x = \dfrac{7}{2}$ and $x = -\dfrac{5}{2}$

> **Investigate further:** The substitution $u = \frac{x+1}{x-2}$ reveals that the second term is $1/u$, so the equation is $u+1/u = 10/3$ — a disguised quadratic in u. Recognise the pattern: whenever you see $f(x)+\frac{1}{f(x)} = k$, substitute $u = f(x)$ to get $u+\frac{1}{u} = k$, i.e. $u^2-ku+1 = 0$.

Section C Substitution & Structure

1. Prove $\dfrac{x}{3}+\dfrac{2y}{3} \geq x^{1/3}y^{2/3}$ for $x, y > 0$ using weighted AM–GM.

Answer: Proved below.

Method: Weighted AM–GM with weights $\frac{1}{3}$ and $\frac{2}{3}$: $\frac{1}{3}\cdot x+\frac{2}{3}\cdot y \geq x^{1/3}y^{2/3}$. $\square$ (This is the standard statement of weighted AM–GM: $\lambda a+(1-\lambda)b \geq a^\lambda b^{1-\lambda}$ for $a, b > 0$, $\lambda \in (0,1)$, here $\lambda = \frac{1}{3}$.)

> **Investigate further:** Weighted AM–GM generalises to: $\lambda_1 a_1+\cdots+\lambda_n a_n \geq a_1^{\lambda_1}\cdots a_n^{\lambda_n}$ where $\lambda_i > 0$ and $\sum \lambda_i = 1$. This is equivalent to the convexity of $-\ln x$. Use it to prove that for $a, b, c > 0$ with $a+2b+3c = 6$: $a\cdot b^2\cdot c^3 \leq \frac{6^6}{6^6} = 1$. (Set $\lambda_1 = \frac{1}{6}, \lambda_2 = \frac{2}{6}, \lambda_3 = \frac{3}{6}$ and apply to a, b, c.)

2. Find all real pairs (x, y) with $x+y = 1$ and $x^4+y^4 = \frac{41}{8}$.

Answer: $\left(\dfrac{3}{2}, -\dfrac{1}{2}\right)$ and $\left(-\dfrac{1}{2}, \dfrac{3}{2}\right)$

Method: Let $p = xy$. Since $x+y = 1$, we have $x^2+y^2 = 1-2p$. Then $x^4+y^4 = (x^2+y^2)^2-2x^2y^2 = (1-2p)^2-2p^2 = 1-4p+2p^2$. Set this equal to $\frac{41}{8}$: $1-4p+2p^2 = \frac{41}{8} \Rightarrow 16p^2-32p-33 = 0$. So $p = \frac{32\pm56}{32}$, hence $p = \frac{11}{4}$ or $p = -\frac{3}{4}$. But $x+y = 1$ gives real roots only if $1-4p \geq 0$, so $p = \frac{11}{4}$ is impossible. Thus $p = -\frac{3}{4}$. Now x, y are roots of $t^2-t-\frac{3}{4} = 0$, i.e. $4t^2-4t-3 = 0$. Hence $t = \frac{3}{2}$ or $t = -\frac{1}{2}$, so $(x, y) = \left(\frac{3}{2}, -\frac{1}{2}\right)$ or $\left(-\frac{1}{2}, \frac{3}{2}\right)$.

> **Investigate further:** Symmetric function method: always reduce to $s = x+y$ and $p = xy$, then express all power sums via Newton's identities. This reduces any symmetric equation in x, y to a polynomial in s and p. The constraint $s = 1$ fixes one variable, leaving a single polynomial in p.

3. Prove $\sqrt{\dfrac{a^2+b^2}{2}} \geq \dfrac{a+b}{2} \geq \sqrt{ab} \geq \dfrac{2ab}{a+b}$ for $a, b > 0$.

Answer: All three inequalities proved below.

Method: (Right) $\sqrt{ab} \geq \frac{2ab}{a+b}$: divide by $\sqrt{ab} > 0$: $1 \geq \frac{2\sqrt{ab}}{a+b}$, i.e. $a + b \geq 2\sqrt{ab}$ — AM–GM. ✓(Middle) $\frac{a+b}{2} \geq \sqrt{ab}$: AM–GM directly. ✓(Left) $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$: square both sides (both positive): $\frac{a^2+b^2}{2} \geq \frac{(a+b)^2}{4} \Leftrightarrow 2(a^2 + b^2) \geq (a + b)^2 \Leftrightarrow (a - b)^2 \geq 0$. ✓□

> **Investigate further:** These are the *Power Mean inequalities*: $M_2 \geq M_1 \geq M_0 \geq M_{-1}$ where $M_r = \left(\frac{a^r+b^r}{2}\right)^{1/r}$ for $r \neq 0$ and $M_0 = \sqrt{ab}$. The chain is $M_2 \geq M_1 \geq M_0 \geq M_{-1}$, i.e. QM $\geq$ AM $\geq$ GM $\geq$ HM. Compute all four for $a = 4, b = 1$.

4. Solve: $27^x + 27^{1-x} = 28$.

Answer: $x = 1$ and $x = 0$

Method: $27^x + \frac{27}{27^x} = 28$. Let $u = 27^x$: $u + \frac{27}{u} = 28 \Rightarrow u^2 - 28u + 27 = (u - 1)(u - 27) = 0$. $u = 1 \Rightarrow x = 0$; $u = 27 \Rightarrow x = 1$.

> **Investigate further:** This is the substitution $u + c/u = k$ (seen in B10 and C of earlier sheets) applied to an exponential equation. The pattern: whenever $a^x + a^{k-x} = c$, let $u = a^x$ to get $u + a^k/u = c$. Here $k = 1$, $a = 27$, $c = 28$.

5. For $a, b, c > 0$ with $a + b + c = 1$, prove $\dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{c} \geq 9$.

Answer: Proved below.

Method: By AM–HM (or AM–GM): $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c} = 9$ by the AM–HM inequality $\frac{n}{\sum \frac{1}{x_i}} \leq \frac{\sum x_i}{n}$, i.e. $\sum \frac{1}{x_i} \geq \frac{n^2}{\sum x_i}$. Alternatively: Cauchy–Schwarz (Engel): $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1^2}{a} + \frac{1^2}{b} + \frac{1^2}{c} \geq \frac{(1+1+1)^2}{a+b+c} = 9$. □

> **Investigate further:** This is the AM–HM inequality for three variables: $\frac{a+b+c}{3} \geq \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$. It is equivalent to $(\sum a)(\sum \frac{1}{a}) \geq n^2$ (by Cauchy–Schwarz). Prove this n-variable version using the Engel form.

6. Evaluate $3^3 + 4^3 + \cdots + 10^3$ using $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Answer: 3016

Method: $\sum_{k=3}^{10} k^3 = \sum_{k=1}^{10} k^3 - 1^3 - 2^3 = \left(\frac{10\times 11}{2}\right)^2 - 1 - 8 = 3025 - 9 = 3016$.

> **Investigate further:** The formula $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2 = \left(\sum_{k=1}^{n} k\right)^2$ is the remarkable *Nicomachus theorem*: the sum of cubes equals the square of the sum of naturals. Prove it by induction, then use it to show $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ — a perfect square for every n.

7. Find all real solutions to $|x^2 - 4| = |x - 2| \cdot |x + 2|$.

Answer: All real x.

Method: $|x^2 - 4| = |(x - 2)(x + 2)| = |x - 2| \cdot |x + 2|$ by the multiplicative property of absolute values: $|ab| = |a||b|$. This holds for all real x, with no restriction. So the solution set is $\mathbb{R}$.

> **Investigate further:** The identity $|ab| = |a||b|$ is fundamental. It follows from $|a| = \sqrt{a^2}$: $|ab| = \sqrt{(ab)^2} = \sqrt{a^2b^2} = |a||b|$. Now solve $|x^2 - 4| = |x - 2| + |x + 2|$ (this is *not* always true — investigate

when it holds and when it fails).

8. Given $x, y > 0$ with $x + y = 4$, find the maximum of x^2y. **Method:** Apply AM–GM to $\frac{x}{2}, \frac{x}{2}, y$: $\frac{x/2+x/2+y}{3} \geq \left(\frac{x^2y}{4}\right)^{1/3}$. Since $x + y = 4$, this gives $\frac{4}{3} \geq \left(\frac{x^2y}{4}\right)^{1/3}$, so $x^2y \leq \frac{256}{27}$. Equality when $x/2 = x/2 = y$, i.e. $x = \frac{8}{3}$, $y = \frac{4}{3}$.

Answer: Maximum is $\dfrac{256}{27}$ at $x = \dfrac{8}{3}$, $y = \dfrac{4}{3}$.

Investigate further: The **splitting trick** for AM–GM: to maximise $x^a y^b$ subject to $x + y = c$, split x into a equal parts and y into b equal parts, then apply AM–GM to $a + b$ quantities. This always gives the maximum at $x/a = y/b$.

Section D Challenge (SMC / BMO1 difficulty)

1. Vieta jumping: Suppose a and b are positive integers with $a^2 + b^2 = k(ab + 1)$. Show $k = 4$ is achieved by $(a, b) = (8, 2)$, and explain the descent step.

Answer: (a) $(a, b) = (8, 2)$ gives $k = 4$. (b) Vieta jumping establishes no new solutions beyond the obvious family.

Method: (a) $a = 8, b = 2$: $64 + 4 = 68 = 4(16 + 1) = 68$. ✓So $k = 4$ is achieved. (b) Fix k and b; consider a as a root of $x^2 - kbx + (b^2 - k) = 0$. The other root a' satisfies $a + a' = kb$ (sum of roots) and $aa' = b^2 - k$ (product). So $a' = kb - a$ is an integer, and $a' \cdot b = b^2 - k$, so $a'^2 + b^2 = k(a'b + 1)$ (verify). If $a > b$, then $a'^2 = k(a'b + 1) - b^2$; since $a' \cdot b = b^2 - k < b^2$ (if $k > 0$), we get $a' < b$. So we can descend: $(a, b) \to (b, a') \to \ldots$ until we reach a base case.

Investigate further: Vieta jumping is a powerful technique for Diophantine equations with symmetric structure. It was used to solve IMO 1988 Problem 6, widely regarded as one of the hardest competition problems ever set. Look up the problem: "$a^2 + b^2 \equiv 0 \pmod{ab + 1}$: prove $\frac{a^2+b^2}{ab+1}$ is a perfect square." The full proof uses Vieta jumping to show the minimum-value solution is $(0, 0)$, giving the value as a perfect square.

2. Find all polynomials $p(x)$ satisfying $(x - 1)p(x + 1) = (x + 2)p(x)$.

Answer: $p(x) = cx(x - 1)(x + 1)$ for any constant c.

Method: $x = 1$: $0 = 3p(1) \Rightarrow p(1) = 0$. $x = -2$: $(-3)p(-1) = 0 \Rightarrow p(-1) = 0$. $x = 0$: $(-1)p(1) = 2p(0) \Rightarrow p(0) = 0$. So x, $(x-1)$, and $(x+1)$ divide p. Try $p(x) = x(x-1)(x+1)$: $(x-1)(x+1)(x)(x+2) = (x + 2)x(x - 1)(x + 1)$. ✓By degree considerations, this is the unique (up to scalar) polynomial solution.

Investigate further: Functional equations for polynomials can be solved by: (1) substituting specific values to find roots, (2) using degree arguments, (3) comparing leading coefficients. This is related to the Gamma function: $\Gamma(x + 1) = x\Gamma(x)$. The polynomial $p(x) = x(x + 1)$ satisfies $(x - 1)p(x + 1) = (x + 2)p(x)$, analogously to Γ.

3. For $a + b + c = 1$, $a, b, c > 0$, find $\min\left(\dfrac{a}{1-a} + \dfrac{b}{1-b} + \dfrac{c}{1-c}\right)$.

Answer: $\dfrac{3}{2}$

Method: $\frac{a}{1-a} = \frac{a}{b+c}$. So the sum is $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b}$. By Nesbitt's inequality: $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$.

Equality at $a = b = c = \frac{1}{3}$.

> **Investigate further: Nesbitt's inequality**: for $a, b, c > 0$, $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$. Proof: add 1 to each fraction: $\frac{a+b+c}{b+c} + \frac{a+b+c}{a+c} + \frac{a+b+c}{a+b} \geq \frac{9}{2}$ by AM–HM on $(b+c), (a+c), (a+b)$. Divide by $a + b + c = 1$. This inequality appears frequently in competition problems.

4. Evaluate $\displaystyle\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}$.

Answer: $\dfrac{2}{3}$

Method: $\frac{n^3-1}{n^3+1} = \frac{(n-1)(n^2+n+1)}{(n+1)(n^2-n+1)}$. The infinite product splits into two:

$\prod_{n=2}^{N} \frac{n-1}{n+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{N-1}{N+1} = \frac{1 \cdot 2}{N(N+1)} \to 0$.

$\prod_{n=2}^{N} \frac{n^2+n+1}{n^2-n+1}$: note $n^2 + n + 1 = (n+1)^2 - (n+1) + 1$ and $n^2 - n + 1 = n^2 - n + 1$. Setting $a_n = n^2 - n + 1$: $a_{n+1} = (n+1)^2 - (n+1) + 1 = n^2 + n + 1$. So $\frac{n^2+n+1}{n^2-n+1} = \frac{a_{n+1}}{a_n}$. This telescopes: $\prod_{n=2}^{N} \frac{a_{n+1}}{a_n} = \frac{a_{N+1}}{a_2} = \frac{N^2+N+1}{3}$.

Combined: $\prod_{n=2}^{N} \frac{n^3-1}{n^3+1} = \frac{1 \cdot 2}{N(N+1)} \cdot \frac{N^2+N+1}{3} = \frac{2(N^2+N+1)}{3N(N+1)}$.

As $N \to \infty$: $\frac{2(N^2+N+1)}{3N^2+3N} \to \frac{2}{3}$. $\square$

> **Investigate further:** The key insight: split the cubic fraction into a product of two ratios, each of which telescopes separately. This technique — factoring into simpler telescoping components — is the standard approach for infinite products. Try $\prod_{n=2}^{\infty} \frac{n^4-1}{n^4+1}$ using the same idea (factor $n^4 \pm 1$ over $\mathbb{Z}[i]$).

5. Find all integer solutions to $x^3 + y^3 = (x + y)^2$.

Answer: $(n, -n)$ for all $n \in \mathbb{Z}$, and $(1, 0), (0, 1), (1, 2), (2, 1), (2, 2)$

Method: Let $s = x+y$ and $p = xy$. $x^3+y^3 = s(s^2-3p) = s^2$. If $s = 0$: $y = -x$ (infinitely many solutions $(n, -n)$). If $s \neq 0$: $s^2-3p = s \Rightarrow p = \frac{s(s-1)}{3}$. For real x, y, discriminant $s^2-4p = s^2 - \frac{4s(s-1)}{3} = \frac{4s-s^2}{3} \geq 0$. Thus $0 < s \leq 4$. Try $s = 1$: $p = 0 \Rightarrow (1, 0), (0, 1)$. Try $s = 2$: $p = 2/3$ (reject). Try $s = 3$: $p = 2 \Rightarrow (1, 2), (2, 1)$. Try $s = 4$: $p = 4 \Rightarrow (2, 2)$.

> **Investigate further:** The Diophantine equation $x^3 + y^3 = (x + y)^2$ connects to elliptic curves. The substitution $s = x + y$, $p = xy$ reduces the problem to a condition on p in terms of s. For fixed s, x and y are roots of $t^2 - st + p = 0$, requiring the discriminant $s^2 - 4p \geq 0$ for real roots and also being a perfect square for integer roots. Work through the case $s = 3$ explicitly.

Top 5 Patterns — Worksheet #7

1. **Weighted AM–GM:** $\lambda a + (1 - \lambda)b \geq a^{\lambda} b^{1-\lambda}$**.** For maximising $x^a y^b$ with $x + y$ fixed, split into a copies of x and b copies of y, apply AM–GM.

2. **Telescoping products via factored ratios.** Write $\frac{n^3 \pm 1}{n^3 \mp 1}$ as a product of two ratios, check if each telescopes independently. The product then collapses to finitely many surviving factors.

3. **Nesbitt's inequality:** $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$**.** This identity arises in disguised form whenever $1 - a = b + c$ (under $a + b + c = 1$). Recognise the substitution $\frac{a}{1-a} = \frac{a}{b+c}$.

4. **$u + c/u = k$ as a disguised quadratic.** Whenever you see $f(x) + \frac{c}{f(x)} = k$, substitute $u = f(x)$ to get $u^2 - ku + c = 0$.

5. **Vieta jumping: the other root is smaller.** For symmetric Diophantine equations, fix one variable, treat the equation as a quadratic in the other, and show the second root gives a smaller positive solution — establishing descent to a base case.

Common Traps — Worksheet #7

1. **Forgetting $s = 0$ solutions in $x^3 + y^3 = (x + y)^2$.** The case $x + y = 0$ gives infinitely many solutions $(-n, n)$ that are easy to miss if you divide by s too quickly.

2. **Splitting infinite products incorrectly.** $\prod(ab) = \prod a \cdot \prod b$ only when both sub-products converge. Check convergence separately for each factor before splitting.

3. **Applying AM–GM without the splitting trick.** $\max(x^2 y)$ with $x + y = 4$ cannot be solved by plain AM–GM on two terms — you must split x into two equal parts to make the inequality sharp.

4. **Confusing Nesbitt with the Cauchy–Schwarz bound.** Cauchy–Schwarz gives $\sum \frac{a}{b+c} \geq \frac{(a+b+c)^2}{\text{something}}$, which can also derive Nesbitt but is less direct. Use the "add 1 to each term" trick for Nesbitt — it's faster.

5. **Stopping Vieta jumping after one descent step.** The power of the method is the infinite descent — you must argue that the process terminates at a base case, not just that one descent step works.

"Do not worry about your difficulties in mathematics. I assure you mine are still greater." — Albert Einstein

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Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths